上海交通大学
SHANGHAI JIAO TONG UNIVERSITY

Carbon Accounting

Lecture 2

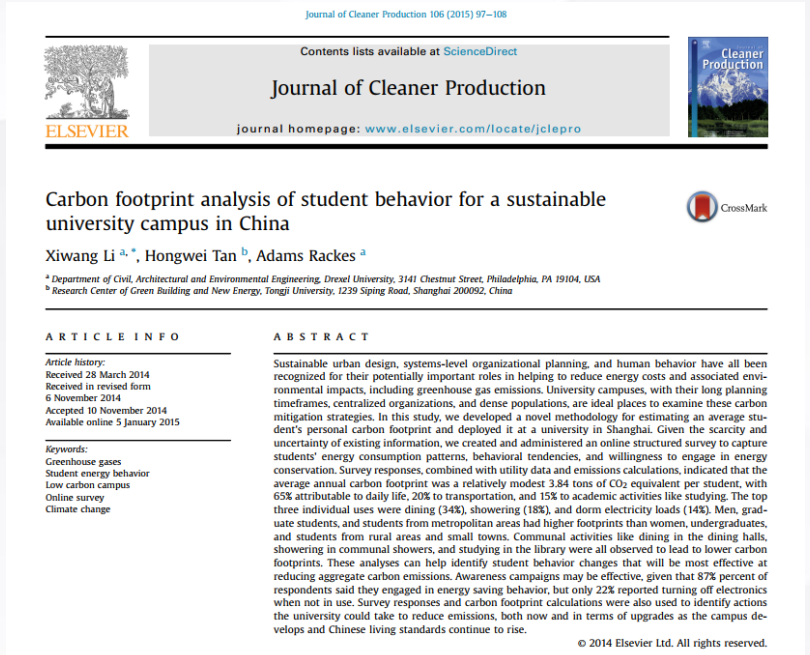
Carbon Accounting: What and Why

Haishan Yu

饮 水 思 源 · 爱 国 荣 校



Tongji University:
3.84 tCO_2e/student/year



Journal of Cleaner Production 106 (2015) 97–108

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Journal of Cleaner Production

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Carbon footprint analysis of student behavior for a sustainable university campus in China

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ABSTRACT

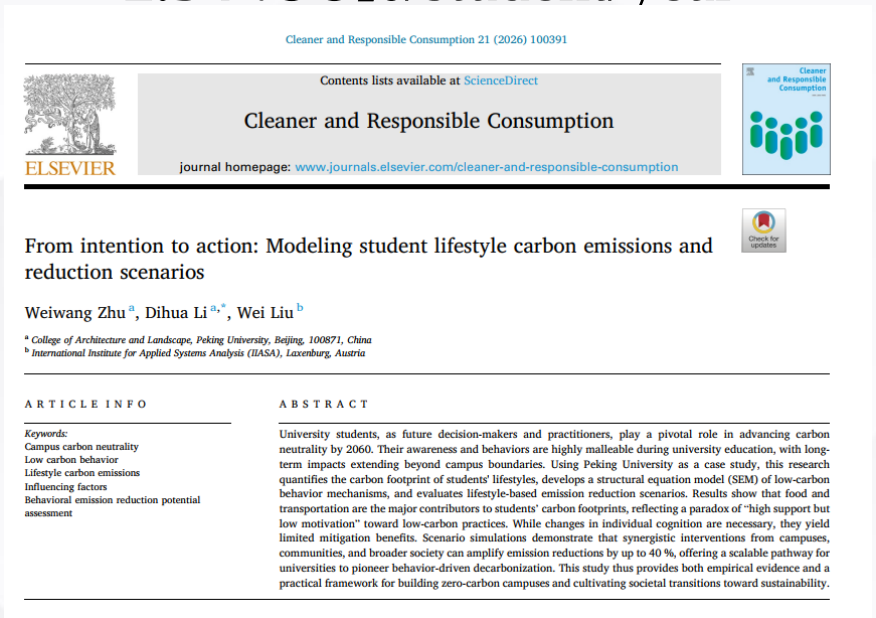
Sustainable urban design, systems-level organizational planning, and human behavior have all been recognized for their potentially important roles in helping to reduce energy costs and associated environmental impacts, including greenhouse gas emissions. University campuses, with their long planning timeframes, centralized organizations, and dense populations, are ideal places to examine these carbon mitigation strategies. In this study, we developed a novel methodology for estimating an average student's personal carbon footprint and deployed it at a university in Shanghai. Given the scarcity and uncertainty of existing information, we created and administered an online structured survey to capture students' energy consumption patterns, behavioral tendencies, and willingness to engage in energy conservation. Survey responses, combined with utility data and emissions calculations, indicated that the average annual carbon footprint was a relatively modest 3.84 tons of CO_2 equivalent per student, with 65% attributable to daily life, 20% to transportation, and 15% to academic activities like studying. The top three individual uses were dining (34%), showering (18%), and dorm electricity loads (14%). Men, graduate students, and students from metropolitan areas had higher footprints than women, undergraduates, and students from rural areas and small towns. Communal activities like dining in the dining halls, showering in communal showers, and studying in the library were all observed to lead to lower carbon footprints. These analyses can help identify student behavior changes that will be most effective at reducing aggregate carbon emissions. Awareness campaigns may be effective, given that 87% percent of respondents said they engaged in energy saving behavior, but only 22% reported turning off electronics when not in use. Survey responses and carbon footprint calculations were also used to identify actions the university could take to reduce emissions, both now and in terms of upgrades as the campus develops and Chinese living standards continue to rise.

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Peking University:
2.54 tCO_2e/student/year



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From intention to action: Modeling student lifestyle carbon emissions and reduction scenarios

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ARTICLE INFO

Keywords:
Campus carbon neutrality
Low carbon behavior
Lifestyle carbon emissions
Influencing factors
Behavioral emission reduction potential assessment

ABSTRACT

University students, as future decision-makers and practitioners, play a pivotal role in advancing carbon neutrality by 2060. Their awareness and behaviors are highly malleable during university education, with long-term impacts extending beyond campus boundaries. Using Peking University as a case study, this research quantifies the carbon footprint of students' lifestyles, develops a structural equation model (SEM) of low-carbon behavior mechanisms, and evaluates lifestyle-based emission reduction scenarios. Results show that food and transportation are the major contributors to students' carbon footprints, reflecting a paradox of "high support but low motivation" toward low-carbon practices. While changes in individual cognition are necessary, they yield limited mitigation benefits. Scenario simulations demonstrate that synergistic interventions from campuses, communities, and broader society can amplify emission reductions by up to 40 %, offering a scalable pathway for universities to pioneer behavior-driven decarbonization. This study thus provides both empirical evidence and a practical framework for building zero-carbon campuses and cultivating societal transitions toward sustainability.

https://doi.org/10.1016/j.clrc.2026.100391

Another question to consider, why would anyone estimate a student's annual emissions in the first place?

Who might find the results useful—and why?

目 录

Contents

01-What is carbon accounting

Meanings, approaches and the focus of this course

02-Why carbon accounting

Users, purposes and information needs

03-Course project launch

From client needs to an initial product concept

04-Wrap-up and next time

Key ideas, preparation and suggested reading

Learning Objectives

1. **Distinguish** broad and narrow understandings of carbon accounting and identify the focus of this course.
2. **Explain** why carbon accounting is needed in different governance and decision-making contexts.
3. **Recognize** major carbon accounting applications and **formulate** initial client needs for an SJTU student carbon footprint tool.

01

What Is Carbon Accounting?

meanings, broad/narrow scope, top-down and bottom-up quantification, and the focus of this course.

Carbon accounting

Carbon Accounting ?

Carbon?

Carbon Peak

Carbon Neutrality

Carbon

Carbon

Carbon

Carbon

Accounting?

Accounting:
https://www.merriam-webster.com/dictionary/accounting：
the system of **recording** and **summarizing** business and financial transactions and **analyzing**, **verifying**, and **reporting** the results
also：the **principles** and **procedures** of this system

核算（会计核算）：
以货币为**计量单位**，运用**专门的**会计方法，对生产经营活动或预算执行过程及其结果进行**连续**、**系统**、**全面**的**记录**、**计算**、**分析**，**定期编制**并提供财务会计**报告**和其他会计资料，为经营决策和宏观管理**提供依据**的一项会计活动。

投票 最多可选1项

设置

In China’s dual carbon goal, is the “carbon” in “carbon peak” and “carbon neutrality” the same concept?

中国的双碳之诺中，碳达峰与碳中和的“碳”是同一个概念吗?

A Yes.

B No.

C Have no idea.

提交

Han – who leads China's leaders group on the efforts to achieve the carbon peaking and neutrality goals, said that "clean and efficient" use of coal was "an important means" to achieve the "dual-carbon" goals during a government conference in March 2022.

韩（他领导着中国实现碳达峰与碳中和目标工作领导小组）在2022年3月的一次政府会议上表示，"清洁高效"利用煤炭是实现"双碳"目标的"重要手段"。

More recently, in November 2023, the NDRC announced that coal power plants will receive payments from 1 January 2024 under a new coal "capacity mechanism". (The UK and several other European countries are among those that have capacity mechanisms.)

最近，在2023年11月，国家发改委宣布，根据新的煤炭"容量机制"，燃煤电厂将从2024年1月1日起获得补贴。（英国和其他几个欧洲国家都设有容量机制。）

'DUAL CARBON' GOALS: "双碳"目标：

"Dual carbon" goals refer to China's two climate goals announced by president Xi Jinping at the 75th session of the United Nations General Assembly in September 2020. President Xi announced that China would reach its carbon emissions peak before 2030 and become "carbon neutral" before 2060. However, it remains unclear if the latter goal refers to the neutrality of carbon dioxide (CO_2) emissions or all greenhouse gas emissions. More broadly speaking, "dual carbon" goals also include China's further climate commitments for 2030 announced by Xi at the Climate Ambition Summit in December 2020. The additional targets include a 65% drop in CO_2 emissions per unit of GDP compared to the 2005 level.

"双碳"目标指的是习近平主席在2020年9月联合国大会第75届会议上宣布的中国两项气候目标。习近平主席宣布，中国将在2030年前实现碳排放量达到峰值，在2060年前实现"碳中和"。然而，后一个目标指的是二氧化碳（CO_2）排放中和还是所有温室气体排放中和，目前尚不清楚。更广泛地说，"双碳"目标还包括习近平主席在2020年12月气候雄心峰会上宣布的中国2030年进一步气候承诺。这些额

This is a comment that was not updated in a timely manner and appears highly imprecise in hindsight.

Despite the lack of rigor in this comment, why did this evaluation initially appear on such an influential consulting platform?

Carbonbrief 是一个聚焦气候变化领域的咨询平台，提供分析报道、客座文章、专家数据库及相关活动资讯。



在联合国气候变化巴库大会世界领导人气候行动峰会上的发言

https://www.fmprc.gov.cn/zyxw/202411/t20241114_11526095.shtml

（2024年11月12日，阿塞拜疆巴库）

丁薛祥

尊敬的各位同事，
女士们，先生们，朋友们：

很高兴作为中国国家主席习近平的特别代表，出席世界领导人气候行动峰会。首先，我谨荣幸地转达习近平主席对阿塞拜疆举办峰会的良好祝愿。习近平主席高度赞赏阿塞拜疆作为COP29主席国为推动全球气候治理作出的积极努力和贡献，并预祝大会取得圆满成功！

今年是《联合国气候变化框架公约》生效实施30周年。站在新起点，我们要携手推动巴库大会取得令人满意的成果，不负国际社会的热切期盼。为此，我谈3点看法和建议。

第一，加强全球气候治理，共同但有区别的责任原则是基石。发达国家应当展现更大雄心和行动，履行率先减排义务，提前碳中和时间，发展中国家也要在能力范围内尽最大努力。中方将提交覆盖全经济范围、包括所有温室气体的2035年国家自主贡献，努力争取2060年前实现碳中和。

中国国家自主贡献（NDC）所规制的温室气体，遵循《巴黎协定》对《联合国气候变化框架公约》的衔接要求，核心涵盖《京都议定书》明确的 6 类温室气体，同时新增三氟化氮（NF_3），具体包括：二氧化碳（CO_2）、甲烷（CH_4）、氧化亚氮（N_2O）、氢氟碳化物（HFCs）、全氟碳化物（PFCs）、六氟化硫（SF_6）以及三氟化氮（NF_3）。

China's 2035 National Determined Contributions (中国2035年国家自主贡献)

1

- reducing economy-wide net **greenhouse gas emissions** by 7–10 percent from peak levels 。

*2015年INDC：二氧化碳排放2030年左右达到峰值并争取尽早达峰；单位国内生产总值二氧化碳排放比2005年 下降60%-65%

*2020年更新的NDC：二氧化碳排放力争 于2030年前达到峰值，努力争取2060年前实现碳中和

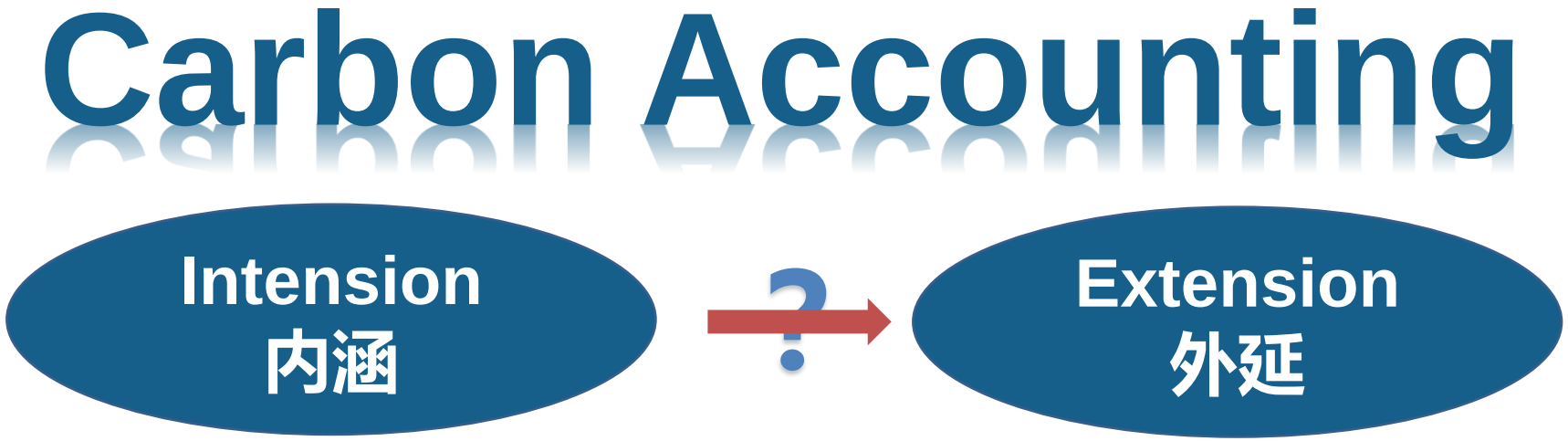
3 Quantitative Targets

- a 30% share of non-fossil fuels in total energy consumption (up from 25% by 2030),
- 3,600 GW of installed wind and solar capacity (up from 1,200 GW by 2030), and
- a forest stock volume of 24 billion cubic metres (from an estimated 18.59 billion cubic metres, or 6 billion cubic metres above 2005 levels).

3 Qualitative Targets

- New energy vehicles have become the mainstream in new vehicle sales.
- The national carbon emission trading market covers major high-emission industries.
- A climate-resilient society has been largely established.

Carbon accounting: intension(内涵) and extension(外延)



Greenhouse Gas Quantification (Broad Sense)
温室气体的量化（广义）

Greenhouse Gas Accounting (Narrow Sense)
温室气体的核算（狭义）

Increasing/Decreasing Intension?

Automobile, Car, Tesla cars, Model 3, Model 3 (2023)



Beef, Livestock, Meat, Food, Goods

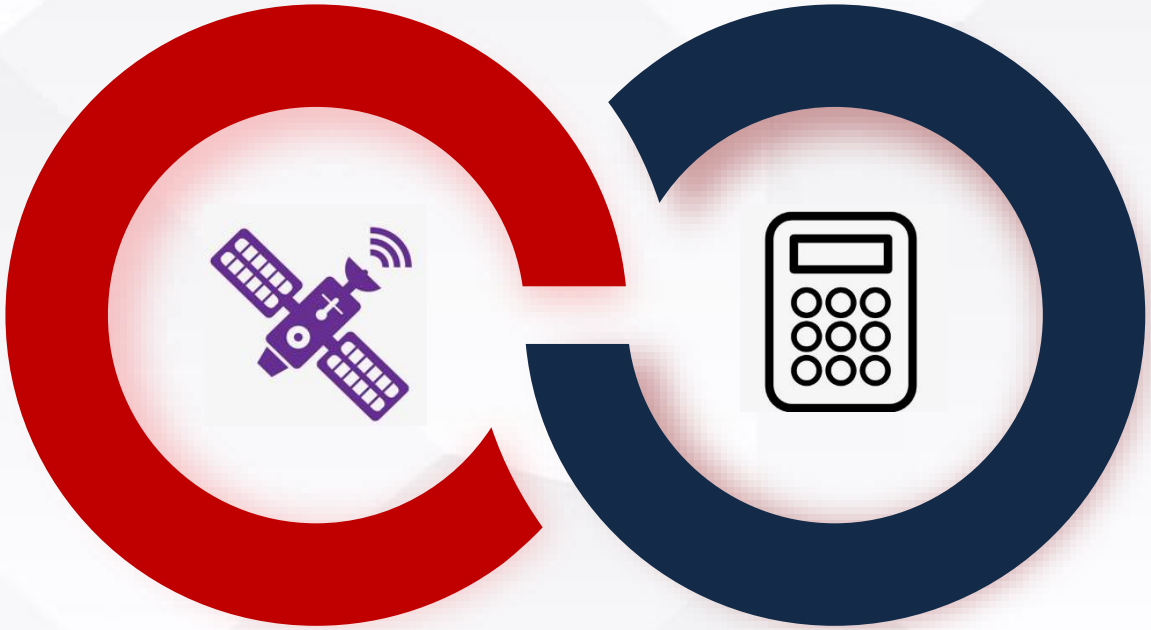


GHG

Measurement/Quantification of GHGs（in the broad sense)

Top Down
（自上而下）

- With remote sensing
  （例如：卫星监测）
- Gridded data
  地图网格化
- Atmospheric CO_2 column measurements
  大气二氧化碳柱浓度
- Inversion
  反演



大数据计算
（大气监测+算法）

核算标准和体系
（MRV）

Bottom Up
（自下而上）

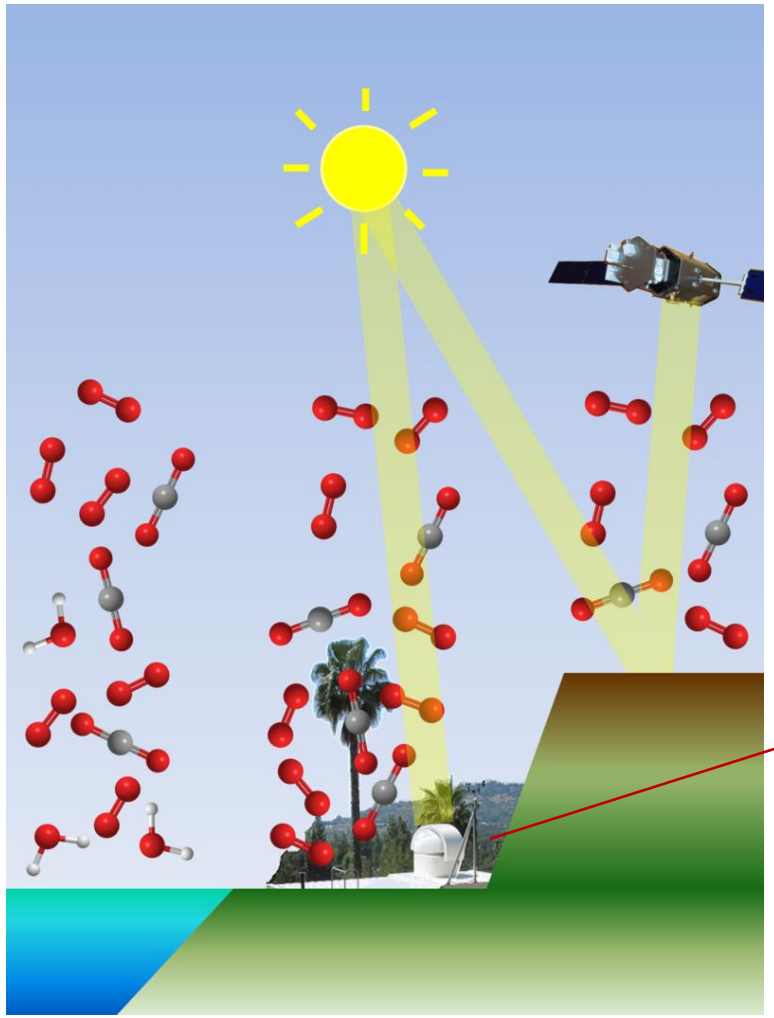
- Clear object(s)
  明确的对象
- Based on (statistical) data
  基于（统计）数据
- Based on calculation
  基于计算：活动水平数据+排放因子；质量平衡法
  Real-time measurement
  基于实时测量： CEMS

Top down GHG Quantification

Top-down Approach

- The ***2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*** introduces a new verification method in its "Quality Assurance and Quality Control" chapter. This method uses a **top-down approach** to invert greenhouse gas (GHG) flux results based on atmospheric concentration data, providing a means to verify and correct traditional bottom-up inventory estimates.
- Key Components:
  - Satellite Remote Sensing Data (Satellite-based Atmospheric GHG concentration)
  - Ground-Based Observations (In situ Atmospheric GHG concentration)
  - Inversion Models (Flux estimation)

Atmospheric GHG concentration - TCCON



Total Carbon Column Observing Network

（总柱碳观测网络）



https://tccondata.org/

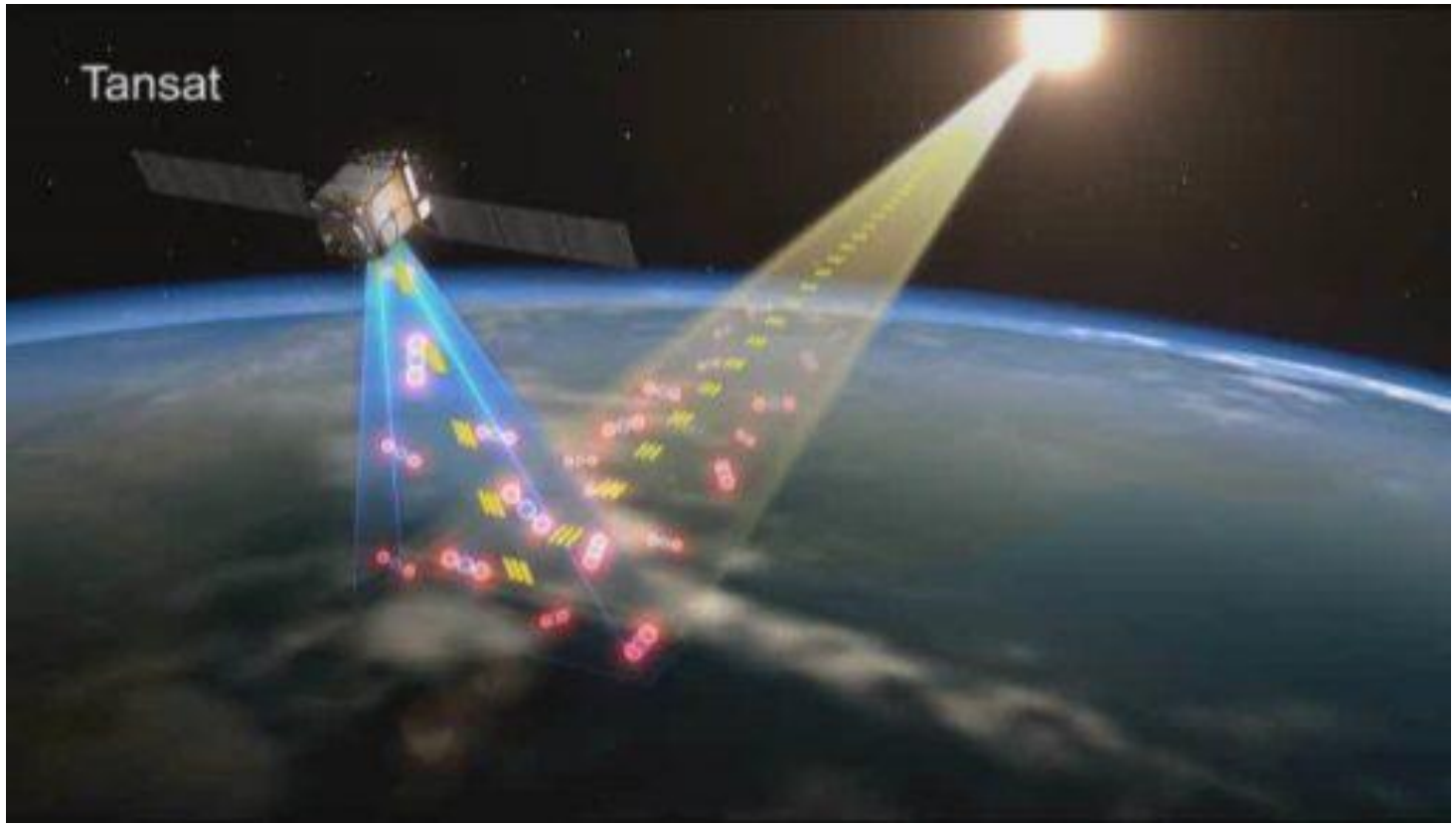
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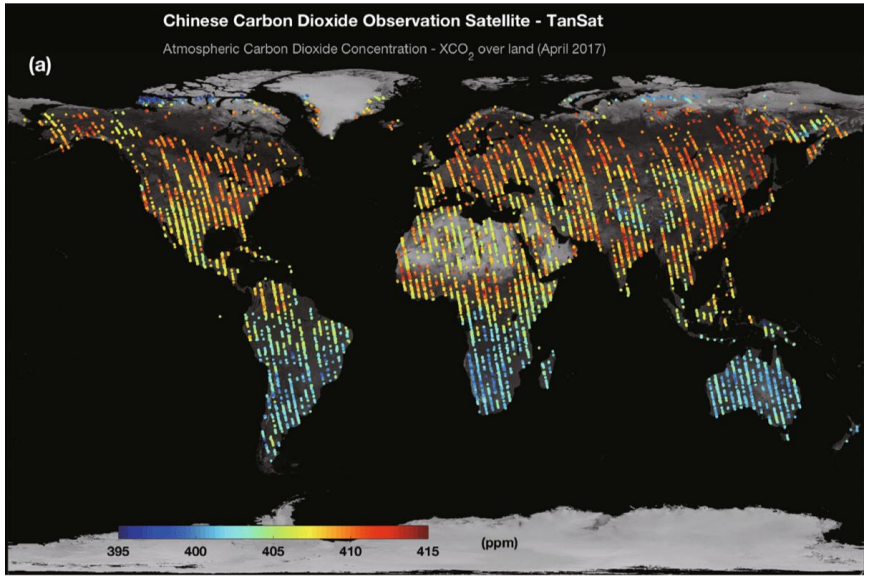
OCO-Orbiting Carbon Observatory



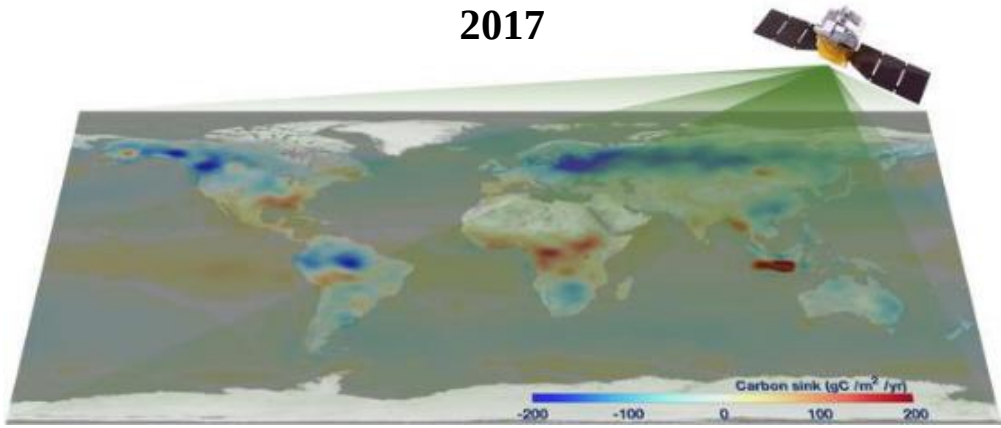
From：任涛

TanSat （碳卫星）





2017



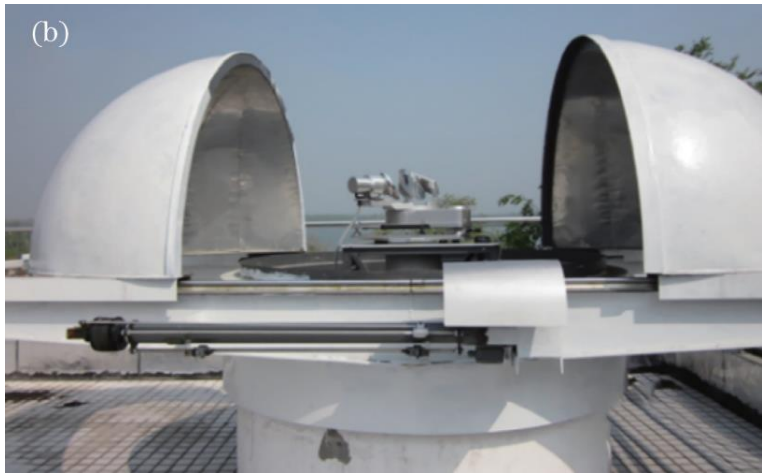
2021

Source: Chinese Academy of Sciences

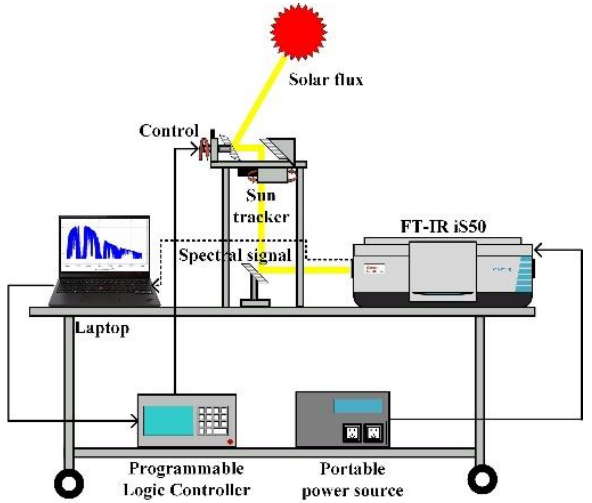
Ground-based CO_2 monitoring platform （地基观测）

TCCON experimental setup





SJTU-LCC experimental setup



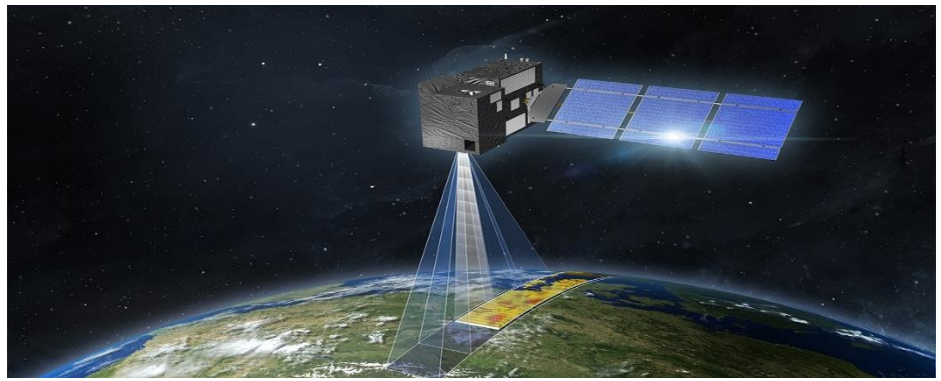


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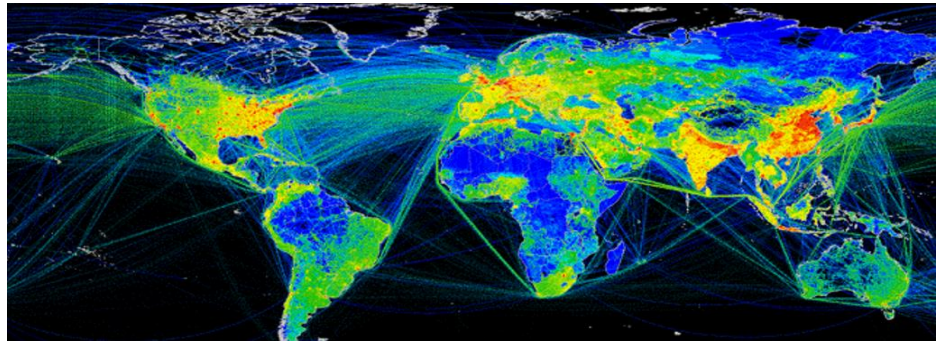
Challenges and opportunities (挑战和机遇)

The next-generation greenhouse gases monitoring satellites mainly address the challenge of improving the spatial and temporal resolutions of observations (进一步提高时空分辨率）

Which will dramatically increase the computational power required for the greenhouse gases retrievals. (计算机算力的挑战）



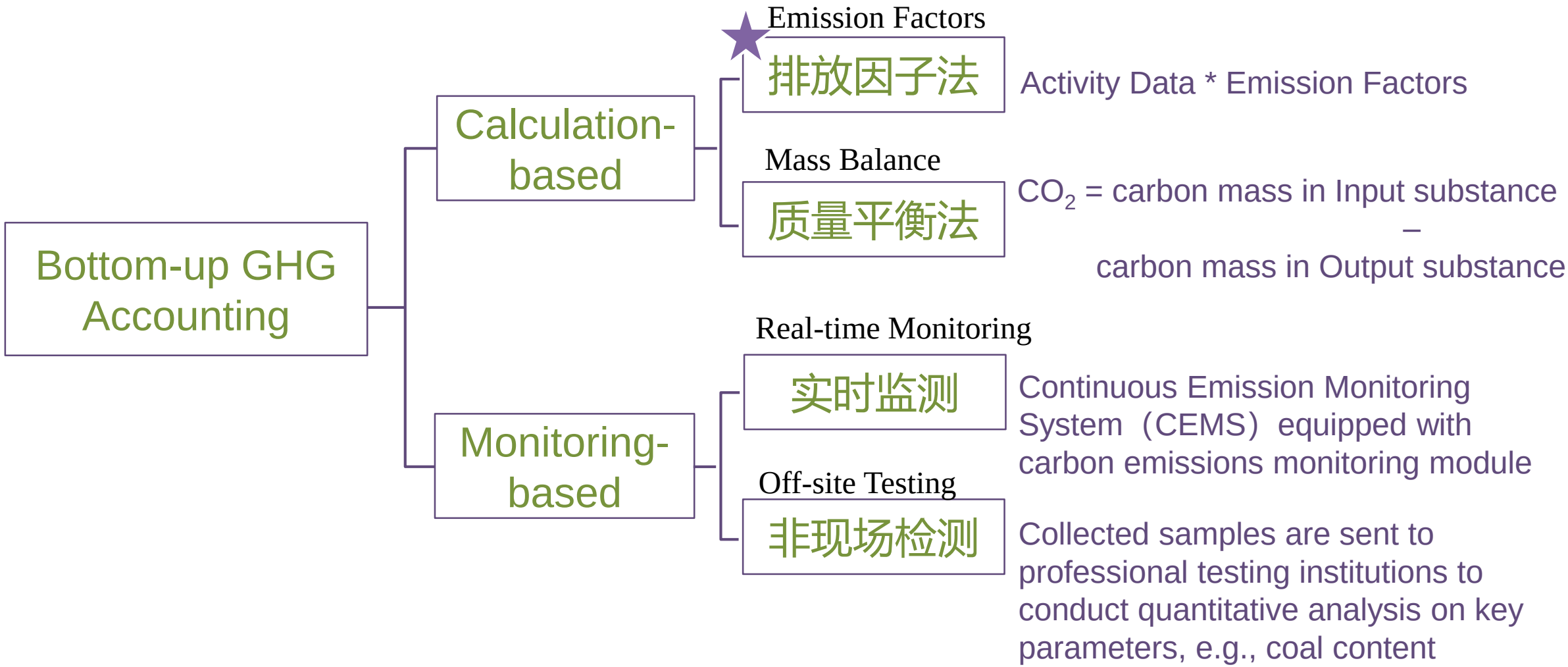




From：任涛

Bottom up GHG Quantification

Bottom-up Approach



Calculation-based GHG Accounting (Extension)

Ascui, F. and Lovell, H. (2011) "As frames collide: Making sense of carbon accounting", *AAAJ* 24 (8): 978-999)

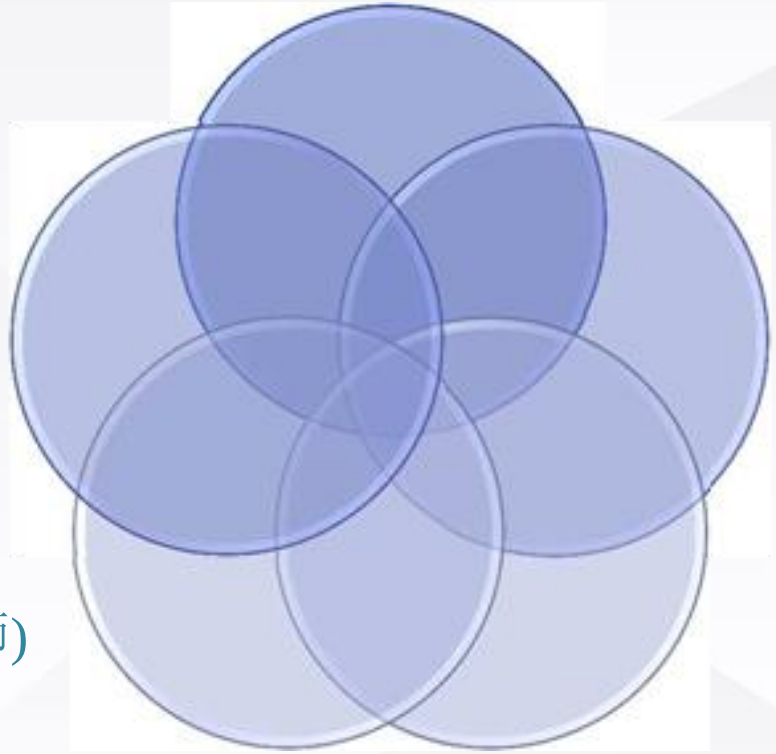
•Estimation (估计) •Calculation (计算) •Measurement (测量) •Monitoring (监测) •Reporting (报告) •Validation (审定) •Verification (核查) •Auditing (审计)	of	carbon carbon dioxide Green-house gas	•emissions to the atmosphere •removals from the atmosphere •emissions rights •emission obligations •emission reductions •legal or financial instruments linked to the above •trades/transactions of any of the above •impacts on climate change •impacts from climate change	at	•Global （全球） •National（国家） •sub-national（地方） •Regional （区域） •civic（城镇） •organisational（组织） •corporate（企业） •project（项目） •installation（设施） •event（活动） •product（产品） •supply chain（供应链）	level for	•Mandatory（强制性） •Voluntary（自愿性）	•Research（研究） •Compliance（合规） •Reporting（报告） •Disclosure（披露） •Bench-marking（基准线设定） •Auditing（审计） •Information（信息） •Marketing（市场） •or other	purposes

Physical

- Scientists （科学家）

Financial

- Accountants （财务会计）



Political

- Policymakers（政策制定者）

Social-environmental

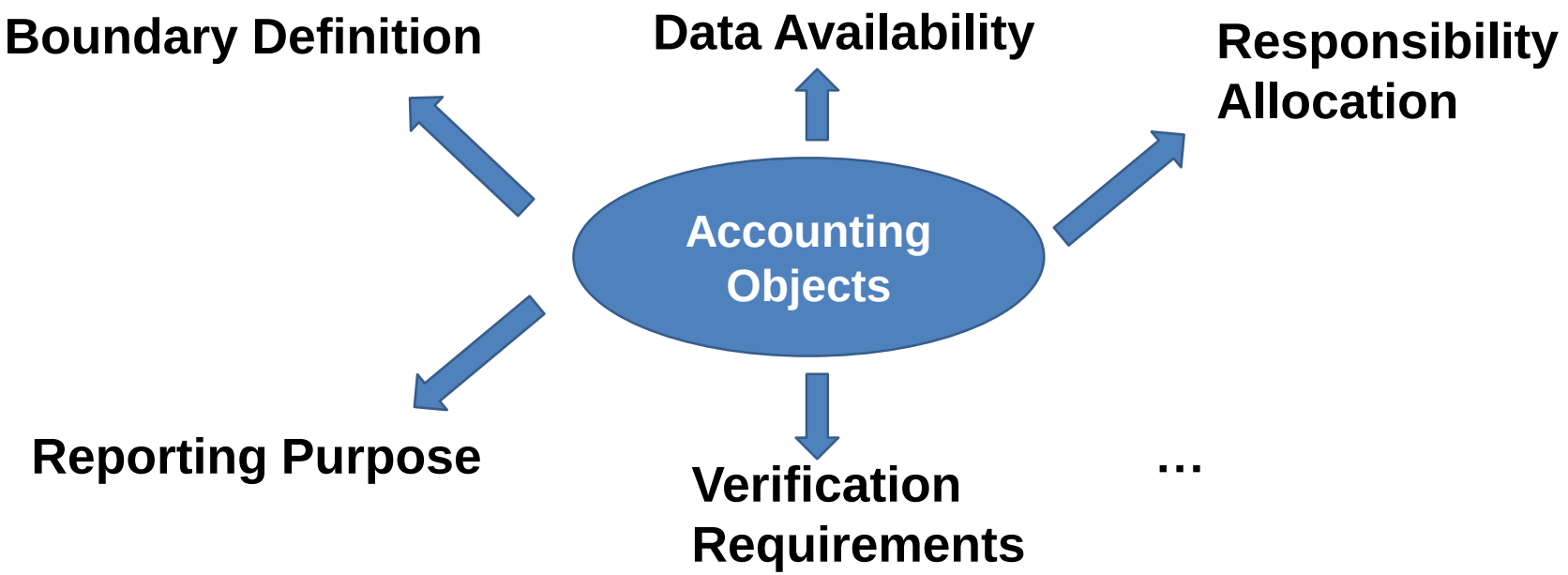
- Carbon managers （碳管理师)
- Footprinting specialists （碳足迹分析师）
- Economists (经济学家）

Market-enabling

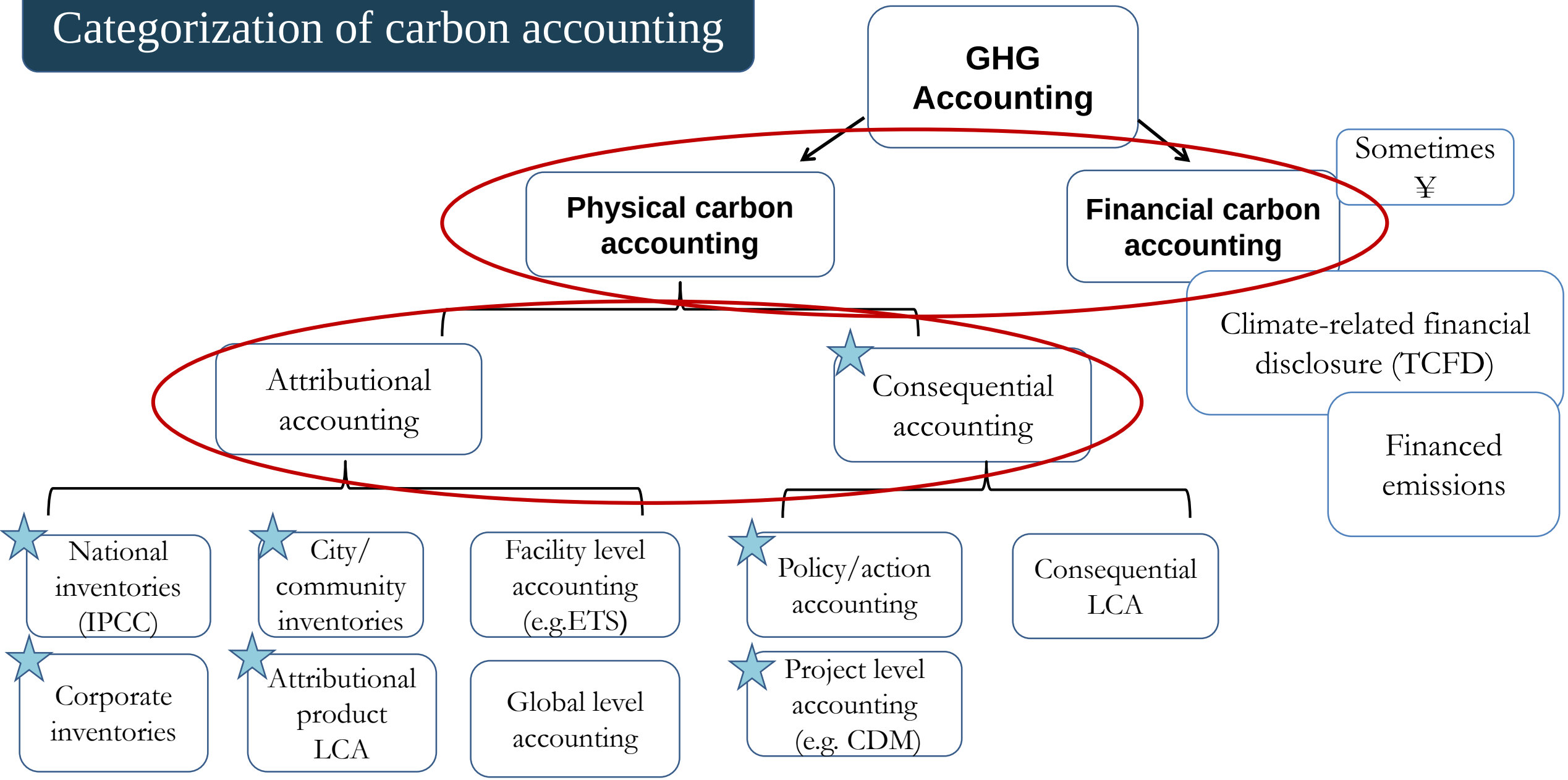
- Carbon traders (碳交易师）
- Project developers (低碳项目开发者）
- GHG auditors （碳审计师）

From Concepts to Framework

- How is carbon accounting organized as a field?
- Different objects usually require different rules because they differ in:



Categorization of carbon accounting

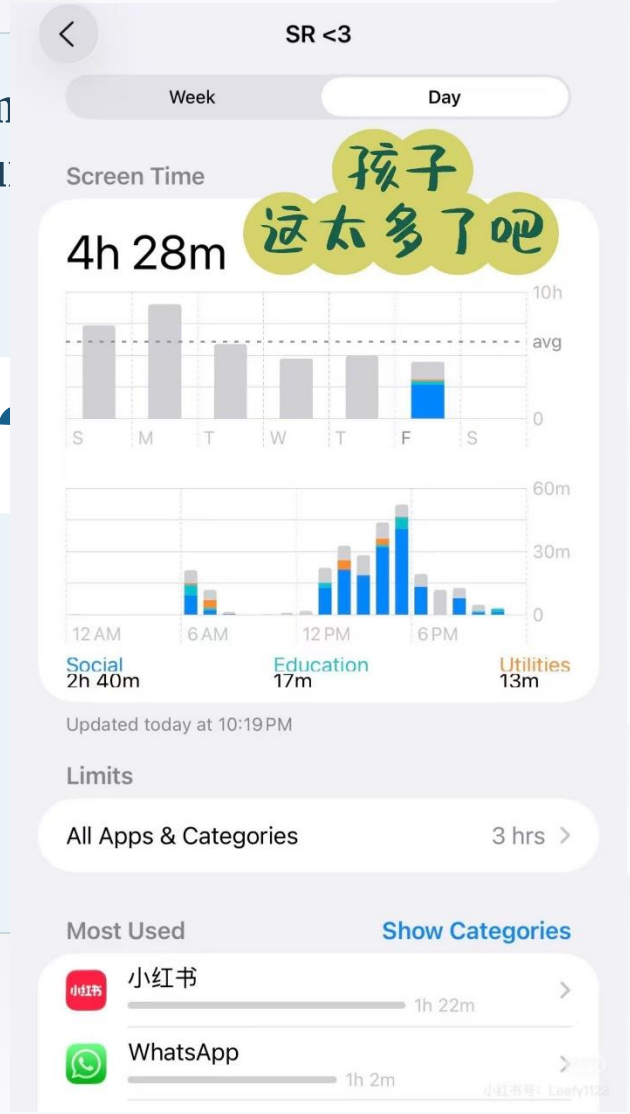


02

Why Carbon Accounting?

A FAMILIAR EXAMPLE

When Does Data Become Useful?



Your phone already records minutes, apps and patterns. **But** a number becomes useful only when someone has a question or decision.

What might you use this information for?

- notice a pattern
- explain a concern
- make a decision
- track a change
- …

Would the daily total alone be enough?

Image source: Xiaohongshu (XHS)@Leafy1128

GROUP DISCUSSION ·5 MINUTES

Now Turn to Emissions Information

U University

A university wants to reduce its climate impact. Students, managers and leaders disagree about priorities and progress.

B Company

An electronics firm faces energy costs, climate commitments and questions from customers, investors and suppliers.

C City

Shanghai is building a low-carbon city — one that requires coordinating transportation, buildings, energy, and waste with limited resources and competing priorities.

N Country

A government develops climate policy, tracks sectors and explains progress internationally while sectors debate responsibility.

Your task: explain who needs emissions information in this situation—and what they need it for.

Use the situation as a starting point. Identify what is at stake, who may use the result, and what could change because of it.

SHARE ·ABOUT 5 MINUTES

Single Round-Robin Add-on Activity



1 First group

Give the main explanation in 30 seconds.

- Who needs the information?
- What is it needed for?

2 Same-context group(s)

Add one missing point or sharpen one link.

Do not repeat what has already been said.

3 Whole class

Listen for a pattern:

What stays similar?
What changes across contexts?

We will rotate by context—not by every group.

The aim is to build a stronger explanation together.

SYNTHESIS –

Across the Four Contexts

What do they have in common?

- Emission information is produced for a particular purpose.
- The accounting subject may not be the only user of the result.
- The result is expected to influence what happens next.
- Useful information must be appropriate for its intended use.

What differ?

- Who uses the result;
- What problem or obligation is being addressed;
- Whether the emphasis is on understanding, reporting, comparison or action;
- What the result needs to represent;
- Who may be made accountable for the emissions.

SYNTHESIS –

What Makes an Emissions Estimate Useful?



Image credit: NOAA

Context

Where is action or accountability needed?

→

User

Who must understand, act or answer?

→

Purpose

What question, claim or decision is at stake?

A carbon account becomes useful when these elements align.

A question we will return to: Is the accounting result fit for its intended use?

Climate Governance Is Also an Information Problem

Climate change is a collective-action problem.

Compare

Are emissions and commitments described on a common basis?

Coordinate

Where should governments, sectors and organizations act?

Track

Are emissions and removals changing over time?

All three require shared, repeatable and reviewable emissions information.

A SHORT HISTORICAL BRIDGE

From Governance Needs to National Inventories

UNFCCC

A shared reporting context

United Nations Framework Convention on Climate Change

→

IPCC Guidelines

Common categories and methods

Inter-governmental Panel on Climate Change

→

National GHG Inventory

Emissions and removals organized nationally

→

Reporting & Tracking

Communicate progress and inform policy

National GHG inventories are among the earliest widely institutionalized carbon-accounting practices.

This course will later return to national inventories—and also examine accounts for cities, organizations, products, projects and individuals.

03

Course Project Launch

Many personal carbon footprint calculators exist — build your own.

碳计算器-碳排放量计算-碳排放计算平台-碳阻迹



https://footprint.kailaimei.com/







www.dotree.com/carbonfootprint/

碳足迹计算器

碳足迹（Carbon Footprint）：一个人在一年内，因衣食住行等活动所消耗的能源，最终转化为大气中二氧化碳(CO_2)的量，用于衡量人类活动对于气候变化的影响。

住宅 | 交通 | 生活 | 绿色习惯

家庭成员: 1 （人）

冬季是否用暖气取暖？：

居住面积: 0 （平方米）

每月用电: 0 （度/月）

每月用气: 0 （立方米/月）(天然气等)

每月用水: 0 （吨/月）

计算结果

您一年排放CO_2: 显示结果 (吨)

Oregon State University Sustainability Office

Carbon Calculator

Sign In

About the Calculator

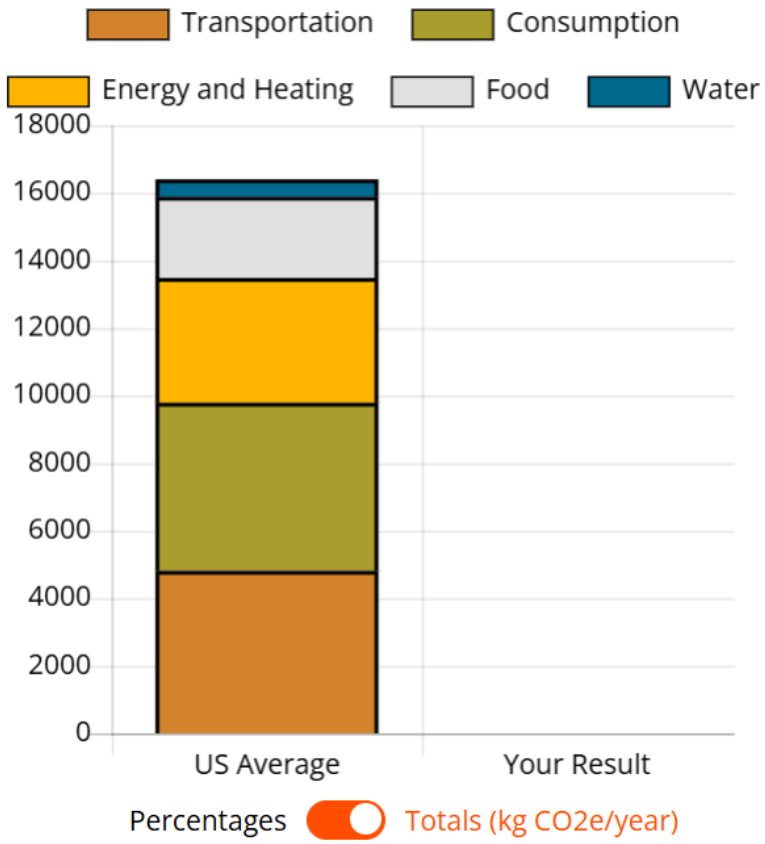
This carbon footprint calculator has been developed to help members of the Oregon State University community understand the connection between their everyday actions and their carbon emissions. This is an important step in Oregon State University's initiative to be carbon neutral by 2025.

To get started, please select which of the following best describes you:

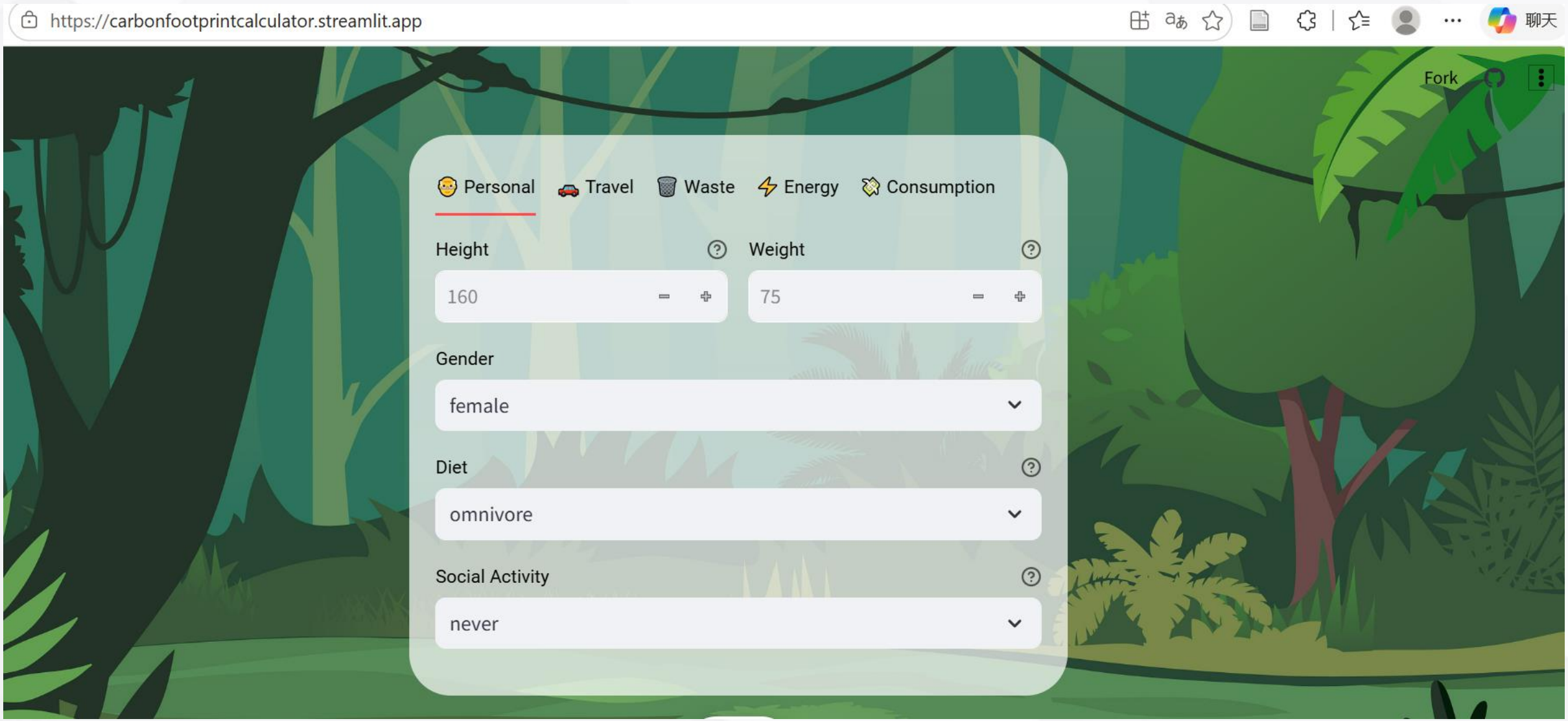
- On Campus Resident Student
- Full Time Commuter Student, Staff, or Faculty
- Part-time Commuter Student, Staff, or Faculty
- Not Affiliated with OSU

Hint: You can use the arrow buttons at the top of the page to navigate between categories.

Results



https://myco2.sustainability.oregonstate.edu/#/



https://carbonfootprintcalculator.streamlit.app/

COURSE PROJECT

Your Design Challenge

Build a web-based personal carbon footprint calculator that a real SJTU student can use.

WHAT THE STUDENT SHOULD BE ABLE TO DO

1. **Enter activities**
   Transport, meals, accommodation, energy consumption, and any other activities of interest
2. **Calculate emissions**
   Computes emissions from the entered activities, reported in kg or tonnes of CO_2e.
3. **Explore the results**
   View the total, browse by category, and compare across activities or against averages.
4. **Get reduction tips**
   Actionable suggestions for lowering emissions, tied to each activity.

YOUR GROUP DECIDES

- **What the tool asks, includes, calculates, and explains**

AI can help build the webpage
— but your group makes and explains the carbon-accounting choices behind the calculator.

PRODUCT REQUIREMENTS

What Will Matter Most?

The project is assessed as a carbon-accounting tool—not as a software competition.

Reasonable

The design fits the intended user and purpose.

Correct

Calculations, units and interpretation are sound and defensible.

Transparent

Boundaries, assumptions, emission factors and limitations are clearly disclosed.

Usable

An SJTU student can enter the required information and obtain clear results that support their decision-making.

A polished interface may improve the experience, but technical complexity is not the deciding criterion.

PROJECT MILESTONES

Course Project Timeline

TODAY

1

Client brief

Shared facts, group assumptions and a mind map

LECTURE 3

2

Prototype 0

First runnable version and the original AI prompt

MID-COURSE

3

Version 1

Test with new accounting concepts and explain revisions

LATER MODULES

4

Version 2

Challenge earlier choices and extend the tests

END OF COURSE

5

Final product

Final tool, version record and project presentation

Keep every version and its prompt log.

Record why the group changes the accounting design.

04

Wrap-Up and Next Time

Wrap-Up and Next Time

SJTU
饮 水 思 源 · 爱 国 荣 校

WHAT

Carbon accounting can be understood broadly or narrowly. This course focuses mainly on calculation-based carbon accounting.

WHY

An estimate is designed for users, purposes and decisions. The same label does not guarantee the same account.

PROJECT

Start from a real student's needs. A sound tool must be reasonable, correct, transparent and usable.

Next time: greenhouse gases, GWP and CO_2e—and your first Prototype 0.

Suggested Readings for Lecture 3

Essential reading:

- US EPA. “Understanding Global Warming Potentials.” access at https://www.epa.gov/ghgemissions/understanding-global-warming-potentials

Recommended reading

- IPCC, 2007: Climate Change 2007: The Physical Science Basis. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change [Solomon, S., D. Qin, M. Manning, Z. Chen, M. Marquis, K.B. Averyt, M. Tignor and H.L. Miller (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, pp.210-213

Project preparation: bring a laptop and be ready to explain the client need your group identified today.





See you next time